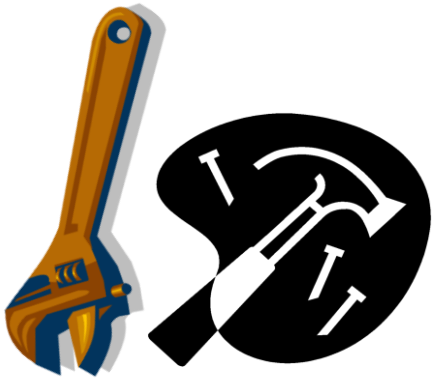
Software Process Models

Major components of system development

Major Components

- Methodology
- Modeling Methods or Techniques
- Tools



Methodology

- A very formal and precise system development process that defines a set of
  - Activities
  - Methods
  - Best practices
  - Deliverables
  - Automated tools

- Used by stakeholders to
  - Develop
  - Continuously improve

} **Information systems and Software**

Methodology

- Provides the framework
- Has a predefined set of steps
- Ensures that systems are built in the most effective way

e.g.
SSADM

Mythology Uses

Modeling Methods or Techniques

Class Diagram,
Use Case Diagrams etc.

Tools

Visual Paradigm,
Rational Rhapsody
Software Ideas Modeler

Tools

Methods

Most Effective
Way of
Building

Software Development Processes

- **Is a set of related activities that lead to the production of a software product.**

Activities may involve:

  - Development of software from scratch (using C++,Java)
  - Development by extending and modify existing software
  - Configuring and integrating off-the-shelf software or components

- Many different Software Processes exist

Examples of activities of a software process

- Software project tracking and control.
- Risk Management.
- Software Quality Assurance.
- Software Configuration Management.
- Work product preparation and production.
- Reusability management, Measurement.

Methodology Vs Process

When building a house,

- the “**process**” - the stages and steps a general contractor takes to get from plan to completed building,
- the “**methodology**” is the skills and capabilities of the individuals responsible for getting the work done (subcontractors, tradespeople, and labourers).

Software Process

- Is there one Process , One Road Map, One predefine template that will take through all the necessary steps from start to finish to make a piece of software?

NO there are many

Common question from a new programmer

Software Development Processes

- All **Processes** must include four main **Phases**
  - Software Specification
  - Software Design and Implementation
  - Software Validation
  - Software evolution
- **Phases** will have **activities**
  - Eg. Establishing a Database
- **Activities** will have **tasks**
  - Eg. Writing a piece of source code

Software Process Models

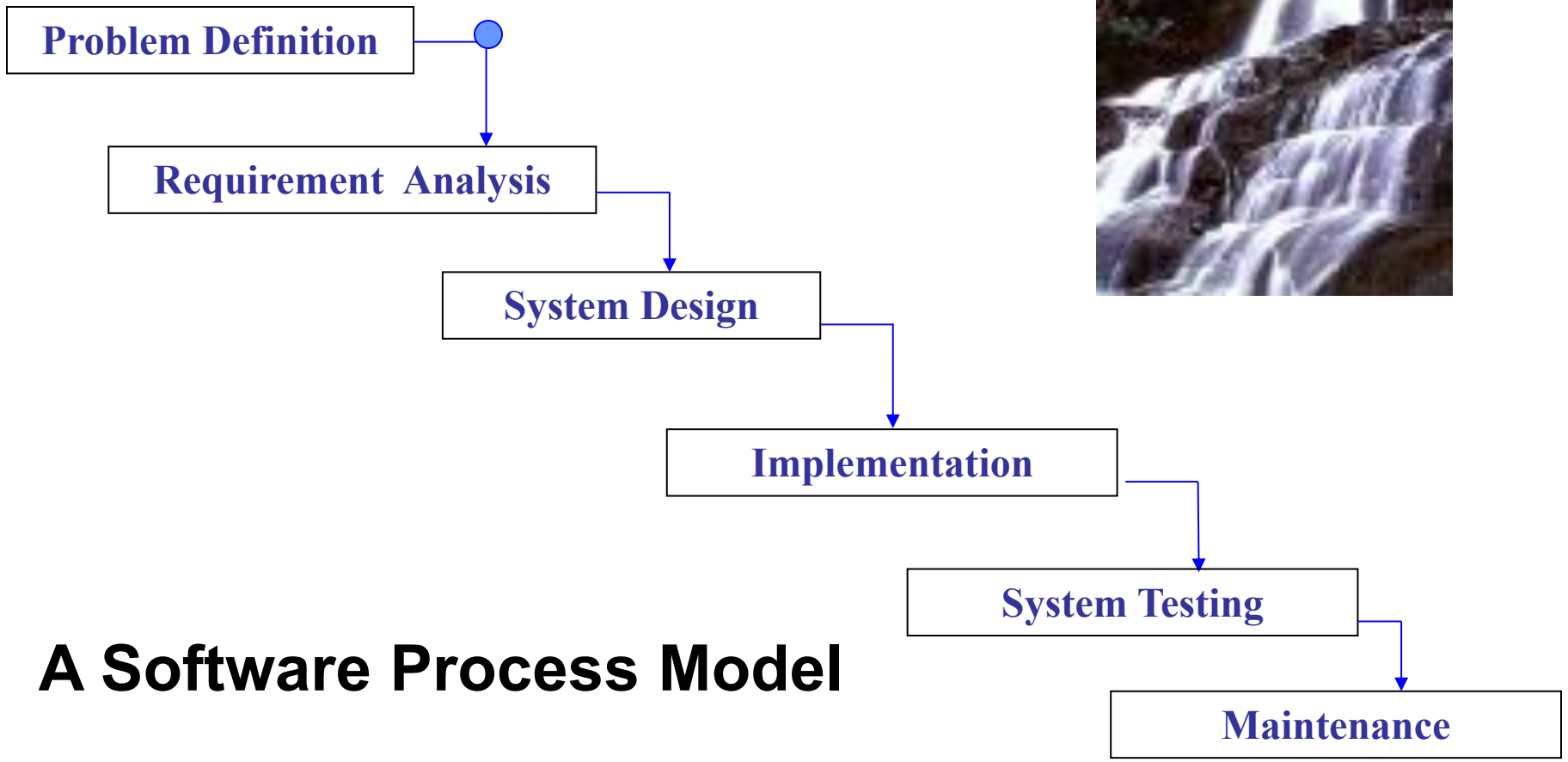
- A simplified representation of a software process

Examples of some popular process models

  - The Waterfall model
  - Incremental development
  - Iterative development
  - Reuse-oriented software engineering

- Software Process models often use together

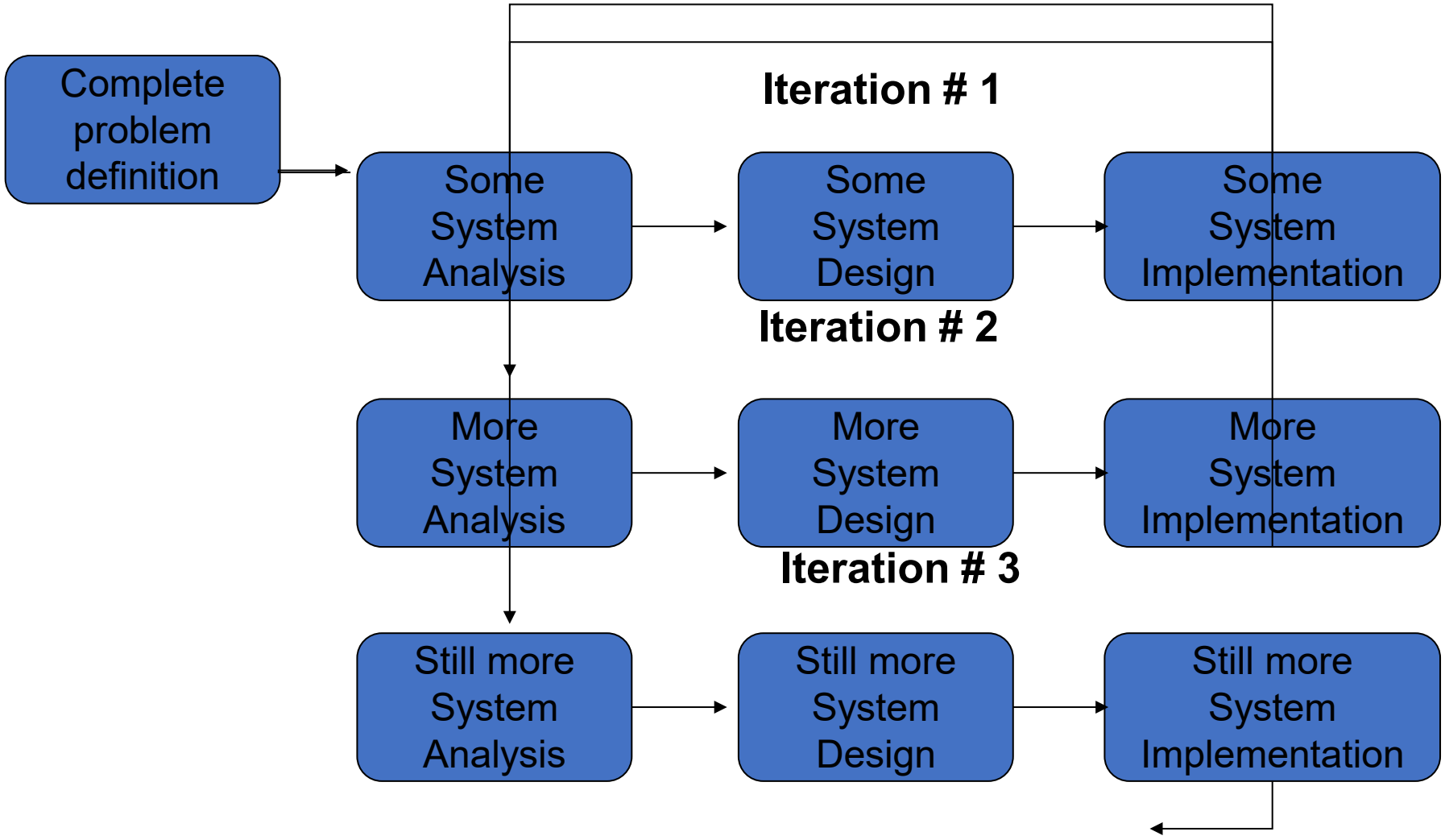
Waterfall Model



A Software Process Model

Iterative development

Systems developed as series of versions

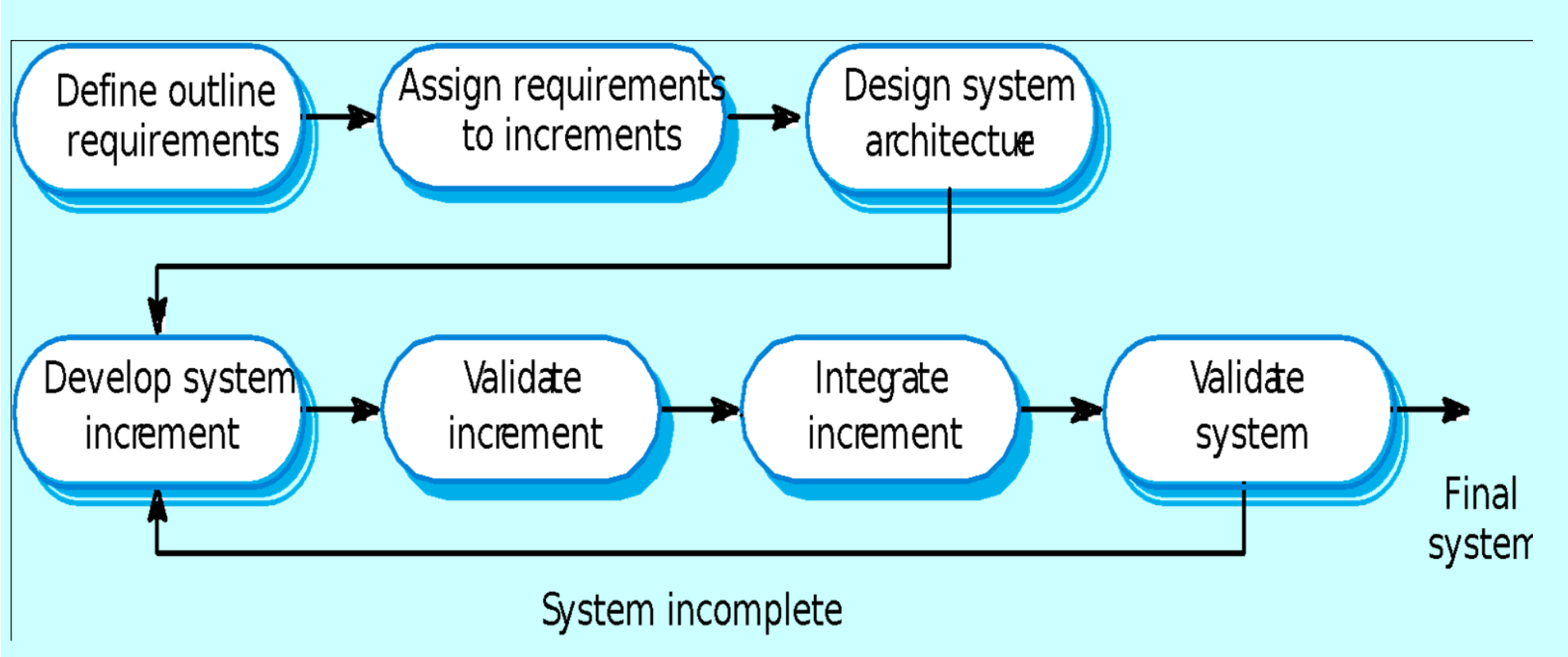


Another Software Process Model

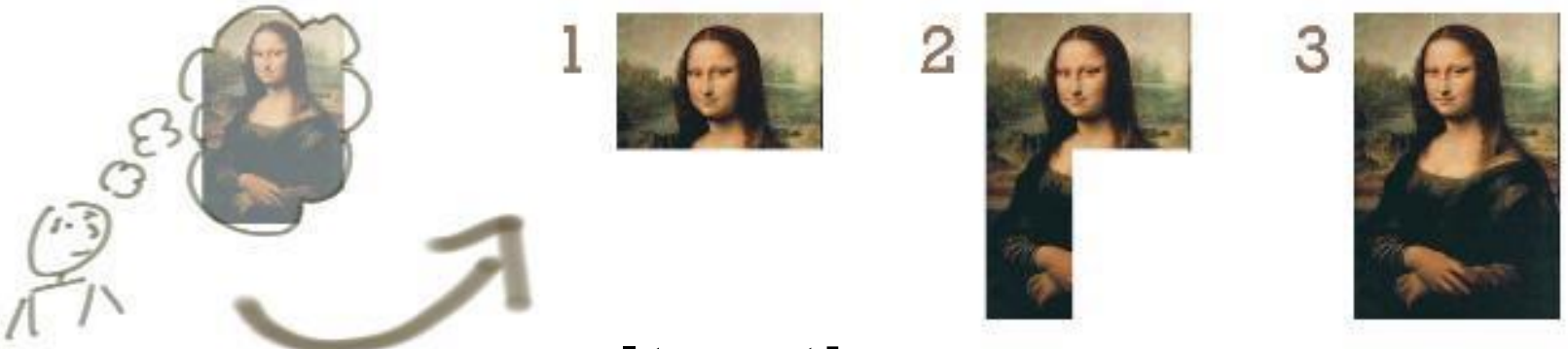
Incremental Development

- Rather than delivering the system as a single delivery, the development and delivery is broken down into **increments** with each increment delivering part of the required functionality.
- User requirements are prioritized and the highest priority requirements are included in early increments.
- Once the development of an increment is started, the requirements are frozen, though requirements for later increments can continue to evolve.

Incremental development model



Incremental



Iterative

INCREMENTAL VS ITERATIVE

Incremental: Build one part completely at a time based on the original idea.

Iterative: Build a rough version, validate this and continue building more quality and functionality in.

Reuse-Oriented Software Engineering

Another software process model

- Approach based on existence of a significant number of reusable components.
- Focus on integrating the components identified in to a system rather than developing them from scratch.

Software Process Models Cont…

- These models often used together
- Combine some of the best features of the process models.
- Sub systems within a large system may be developed using different approaches.